■ Green Up the Roof!

DIADEM®

DiaSafe® Line

EN Permanent anchor devices

EN Installation Guide

Introduction – General description

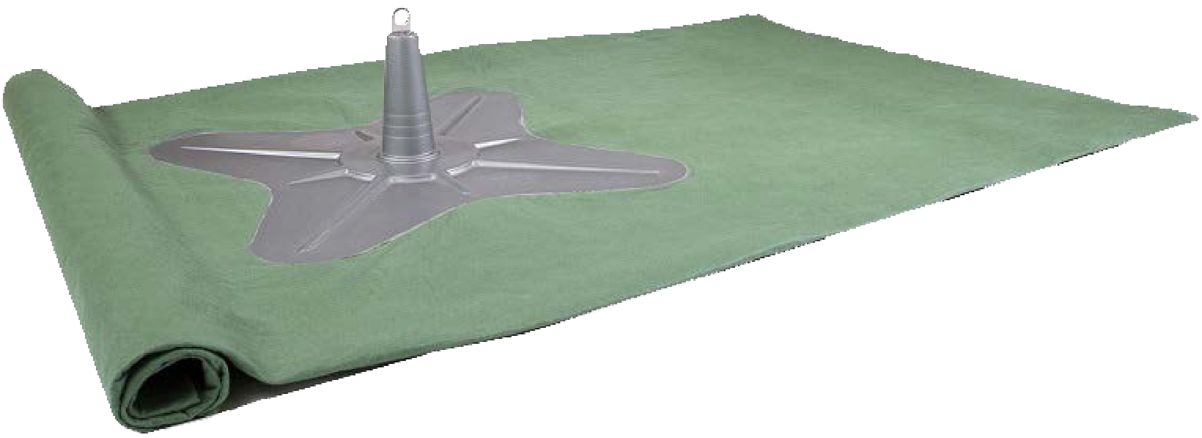
Line wire-rope system

DiaSafe® Line

The **DiaSafe® Line** was developed as a horizontal wire-rope safety technical system, which can be used in full length without disconnecting if using a traveller.

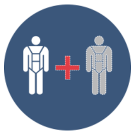
by using a traveller
on the entire system
for the simultaneous secure of 1+1 persons,

by connecting a carabiner
in every second post space
for the simultaneous secure of 1+1 persons



System design, components

DiaSafe® Line wire rope system



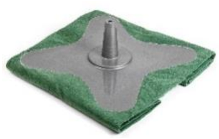
Post structure: DS post + DS amoeba-shaped damping plate
with ballasting mat (3 x 3 m)
Properties: no need to break through the layering sequence of
the roof to perform the installation
Load direction: 360° (horizontal)
Material: stainless steel 1.4301
stainless steel 1.4408
glass-fibre reinforced plastic (amoeba-shaped damping plate)
Fixation: specified ballast material (detailed in Point 6)
Min. distance of posts: 1.5 m
Optimal distance of posts: 7.5 m (max. 8 m)
Post height: 300 mm

DiaSafe® Line / Wire-rope system components

DS Post 2.0
Product No.: 100630
Material: stainless steel 1.4301
Size: Ø250 mm x 285 mm

DS system holder ring (head plate) rotated
Product No.: 100593
Material: stainless steel 1.4408 / surface electropolished
Size: Ø50 x 8 mm

DiaSafe 21 head Kit
Product No.: 100595
Material: stainless steel 1.4408 / surface electropolished
Included: M12 nut
flat washer DIN 913
grub screw M8 x 8 mm



DS Amoeba-shaped damping plate with ballasting mat (3 x 3 m)
Product No.: 100560
Material: glass-fibre reinforced plastic und polypropylene
Size: 3 x 3 m

DiaSafe Loop
Product No.: 100596
Material: stainless steel 1.4408 electropolished
Size: 29 x 29 x 29 mm

DS DiaGlider-Fix traveller (without carabiner)
Product No.: 100471
Material: stainless steel 1.4542
Application: Placed on wire, not detachable.

DS Holder head Kit (for system beginning, ending and T-branching)
Product No.: 130942
Material: stainless steel 1.4301
Included: M12 nut, spring washer

DS Multi Cable Thimble
Product No.: 100279
Material: stainless steel 1.4404
Size: 58 x 38 mm

DS Stainless steel wire-rope
Product No.: 100268
Material: stainless steel 1.4404
Diameter: Ø8 mm (7 × 19)
Tensile strength: F = 33.4 kN

DS Wire-rope terminating shrinkable tube
Product No.: 090845
Size: Ø9 mm

iaSafe® systems accessories

DS fall arrest mat
Product No.: 320324
Material: polypropylene

DS signal adapter
Product No.: 100373
Material: fibreglas reinforced polyester

DS column ring (head plate) turned
Product No.: 100616
Material: stainless steel 1.4301
Size: Ø50 x 8 mm

MAS HA4 rope
Material: 16 mm twisted rope, polyester

Product No.	Length	Carabiners
130981	13 m	with 1 carabiner
130982	16 m	with 2 carabiners
130983	20 m	with 2 carabiners
130984	23 m	with 2 carabiners
130985	25 m	with 2 carabiners
130986	30 m	with 3 carabiners

EN Safety instructions

General information

- The system may only be installed by appropriately trained, competent professionals who are familiar with the safety system.
- The installer of the system must be familiar and comply with the local, and labour safety regulations.
- The plan of the safety system (or a sketch of the rooftop) must be available at the access point to the roof safety system.
- On slanted roofs, the proper snow-catching features must be added to prevent ice or snow from causing damage in the system while sliding from the roof surface.

Safe installation

- Stainless steel may not come into contact with sanding dust or steel tools, as this can lead to corrosion.
- Proper fastening of the safety system must be documented with fastening reports and photos of the respective circumstances of installation.
- The installer must ensure that the substructure is appropriate for the fastening of the anchoring materials. In case of doubt, consult a structural engineer.
- The distance of the safety system from the roof edge is to be defined according to local regulations standards and depending on roof geometry.

EN Mounting surface

The prerequisite for proper and professional installation is a statically sound construction and the use of the original fastening components listed in the Technical Manual.

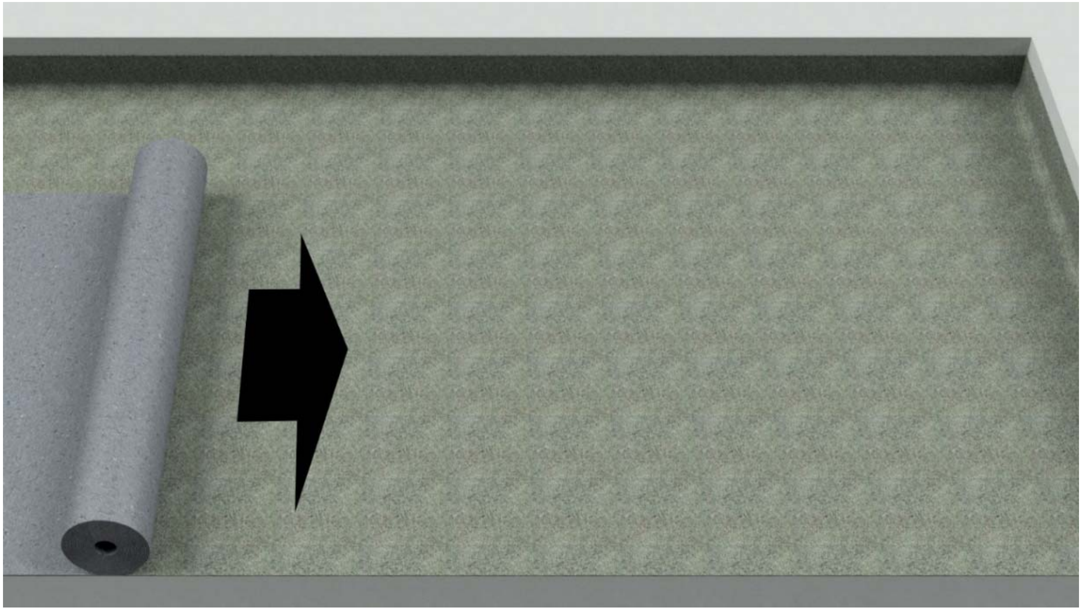
Danger of death during installation!

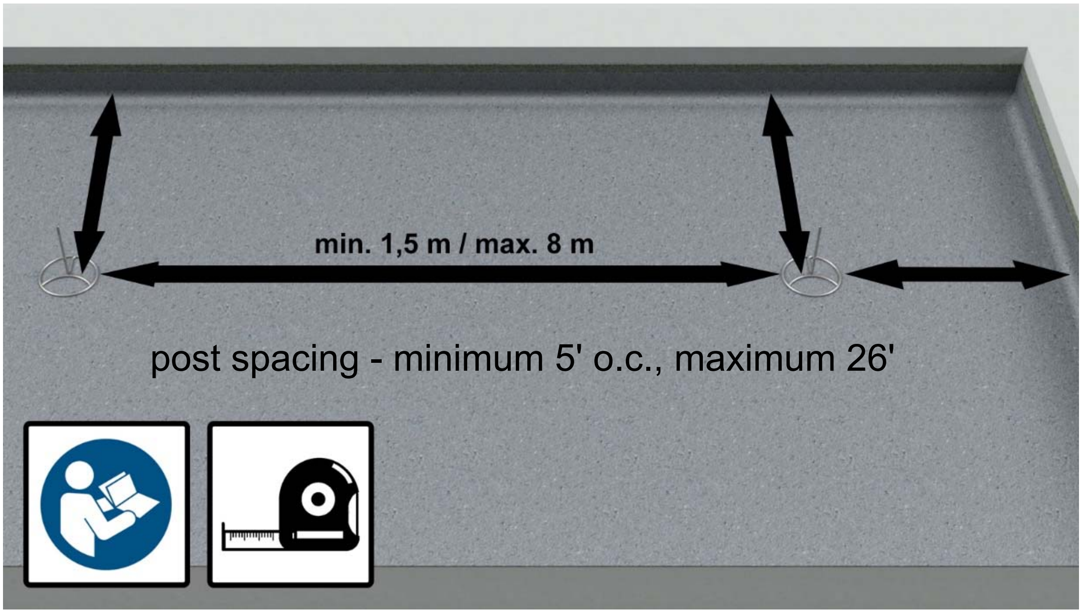
- The fall protection system may only be installed on a statically load-bearing substructure!
- Do not install in streed, blinding concrete, sloping concrete, etc.!
- The roof structure must be statically inspected!
- The technical specifications of the Technical Manual must be completely observed!
- In case of the use of fastening elements, anchors other than those described in the Technical Manual, the responsibility lies with the installer.

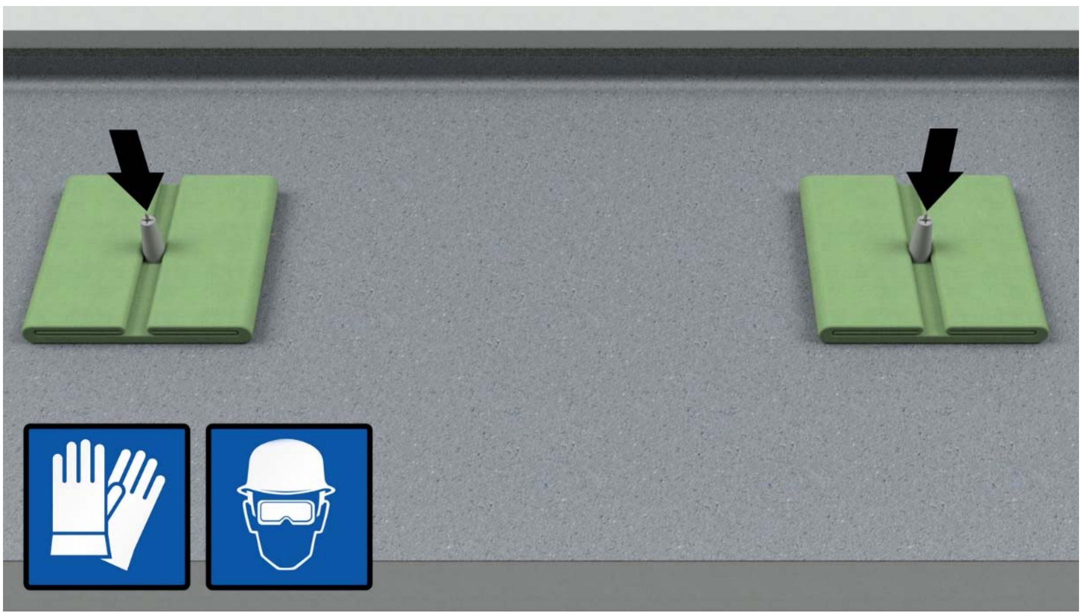
Installation

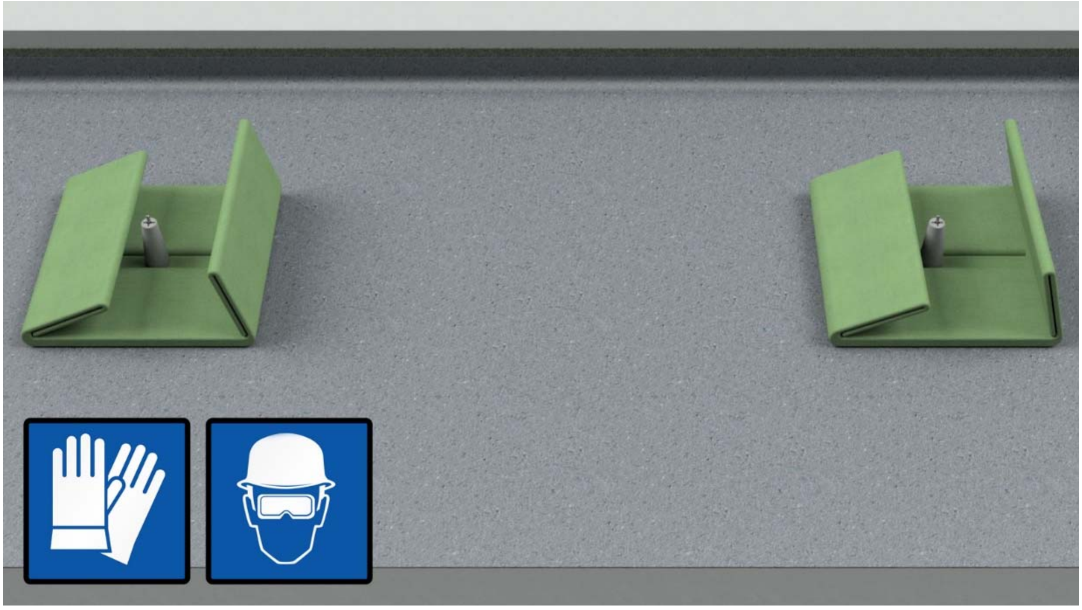
Installation tools

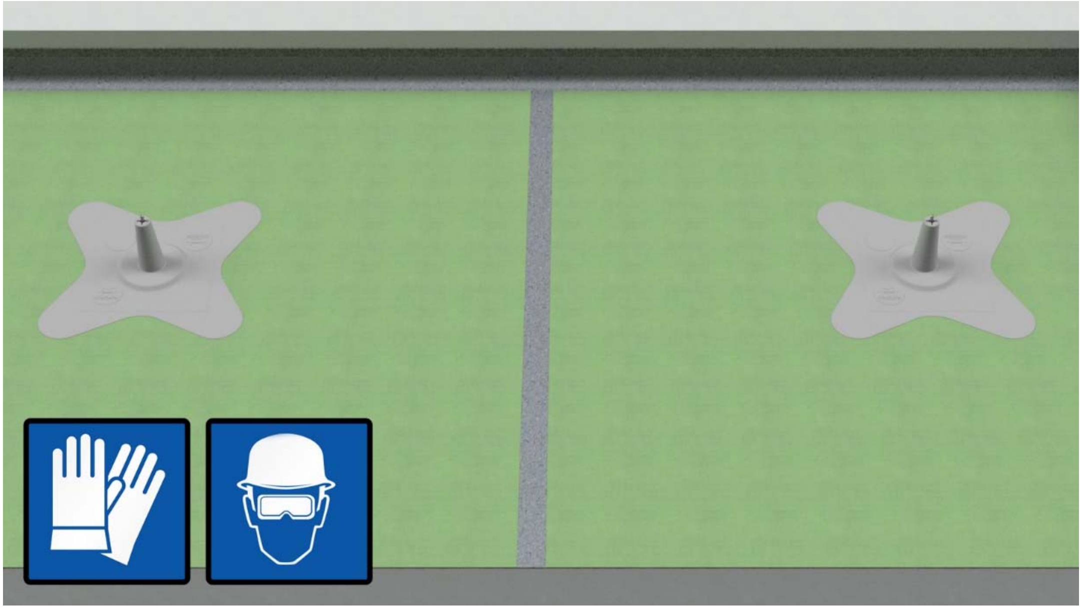
Anchor post installation

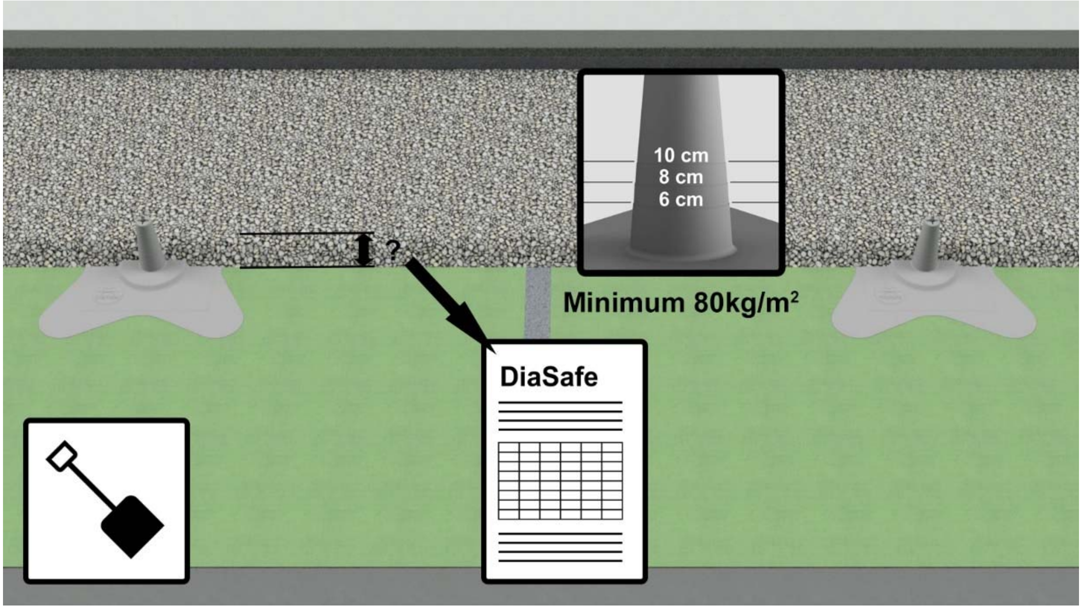




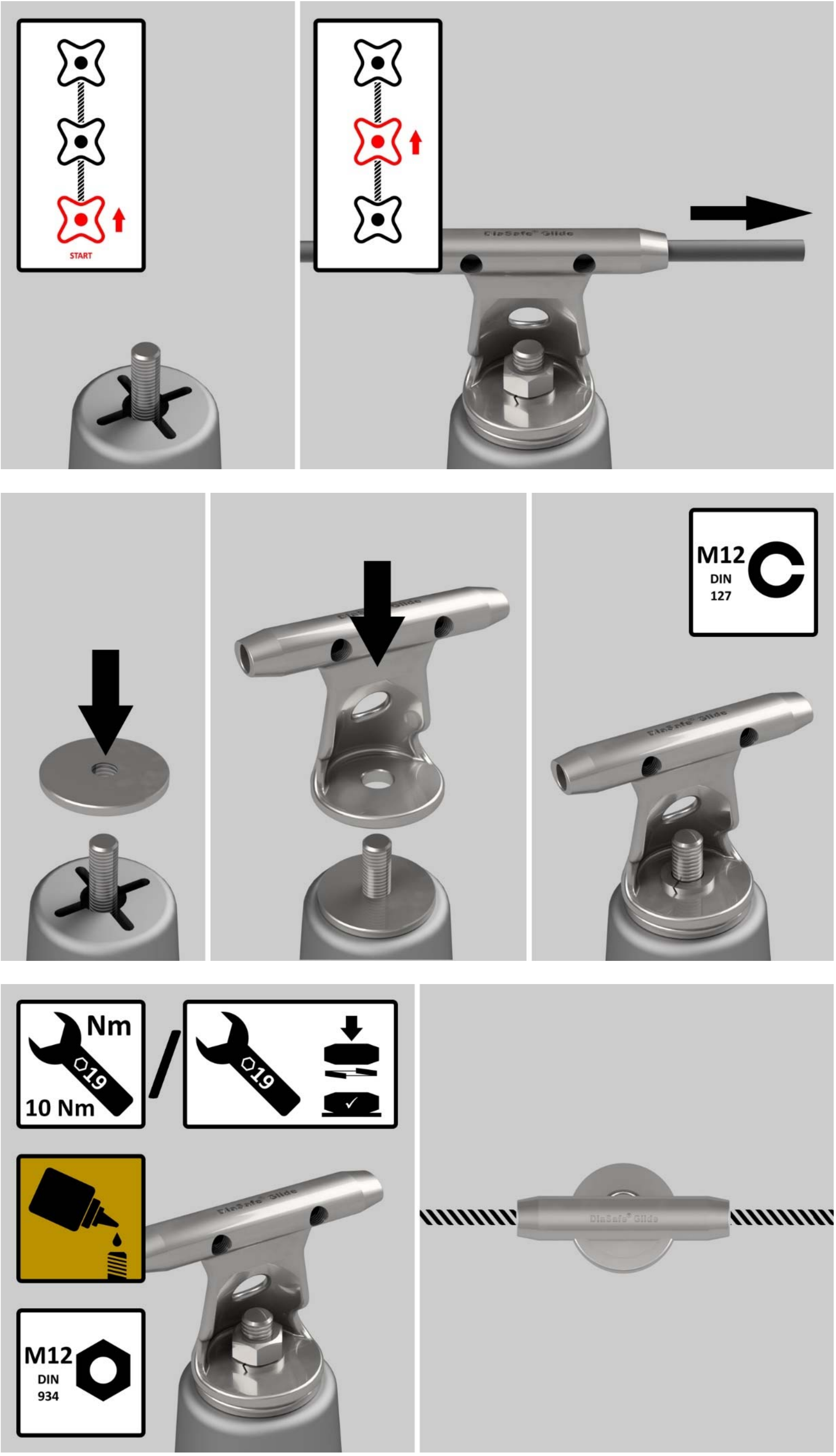


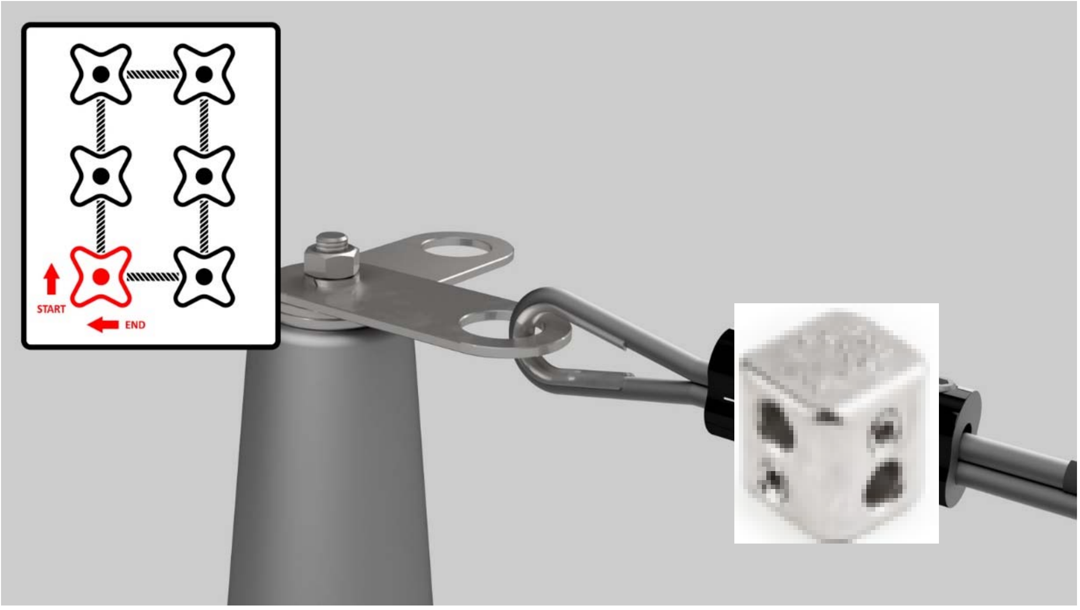


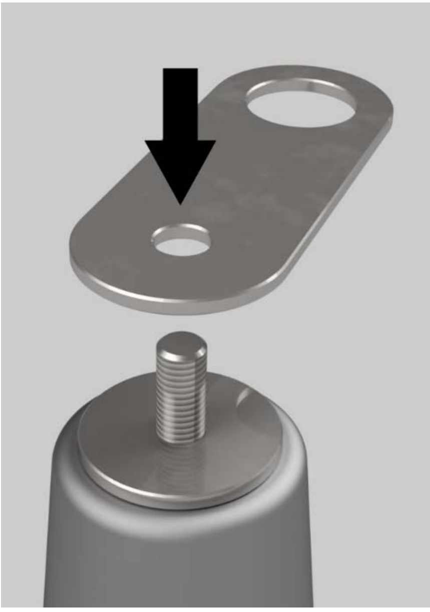


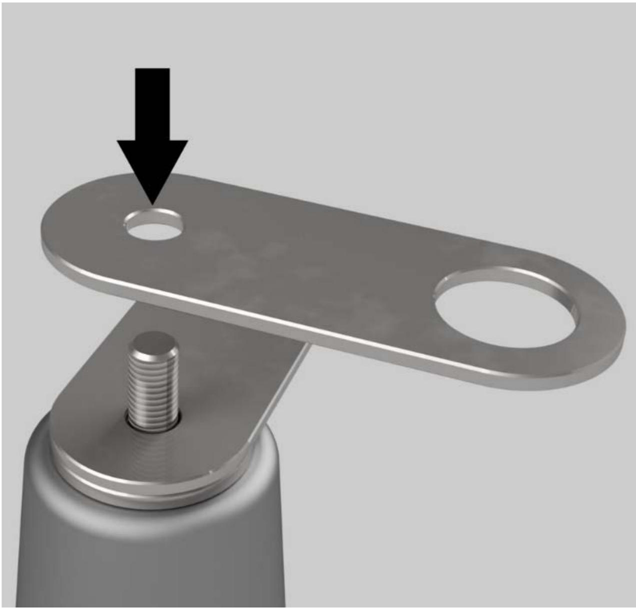

?
10 cm
8 cm
6 cm
Minimum 80kg/m²
DiaSafe

Open line system installation (with start-and-end posts)

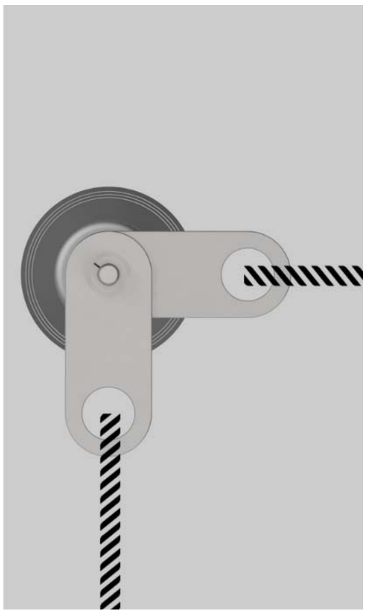


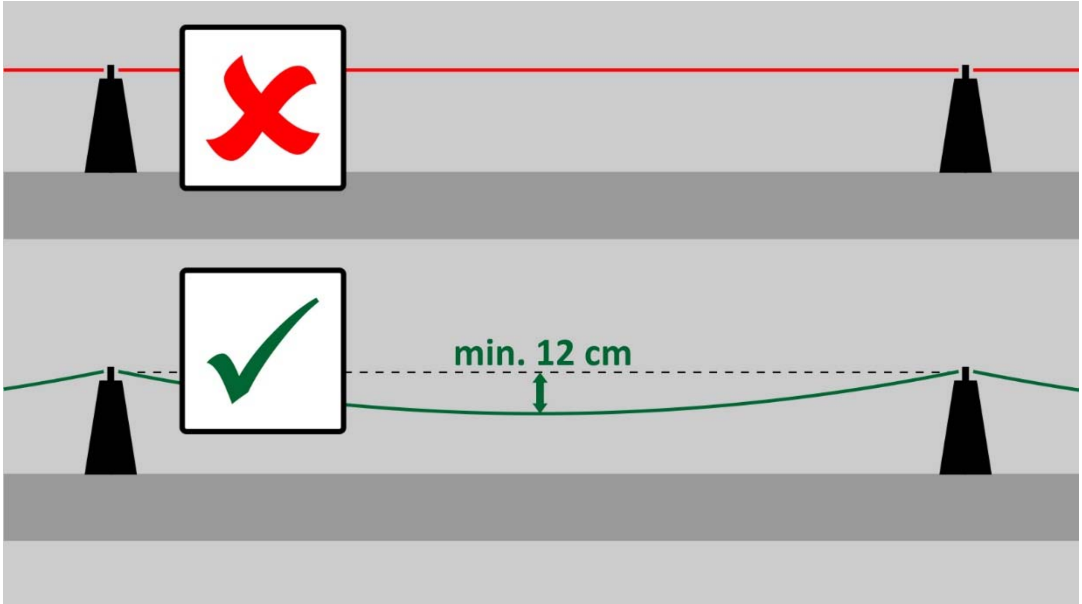

START
END

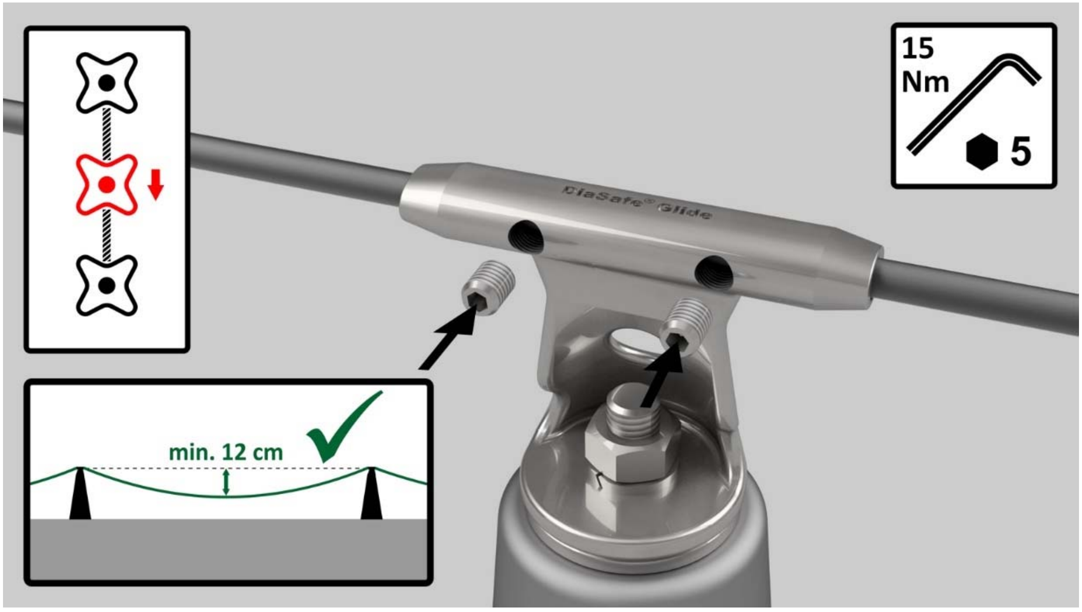





M12
DIN
127

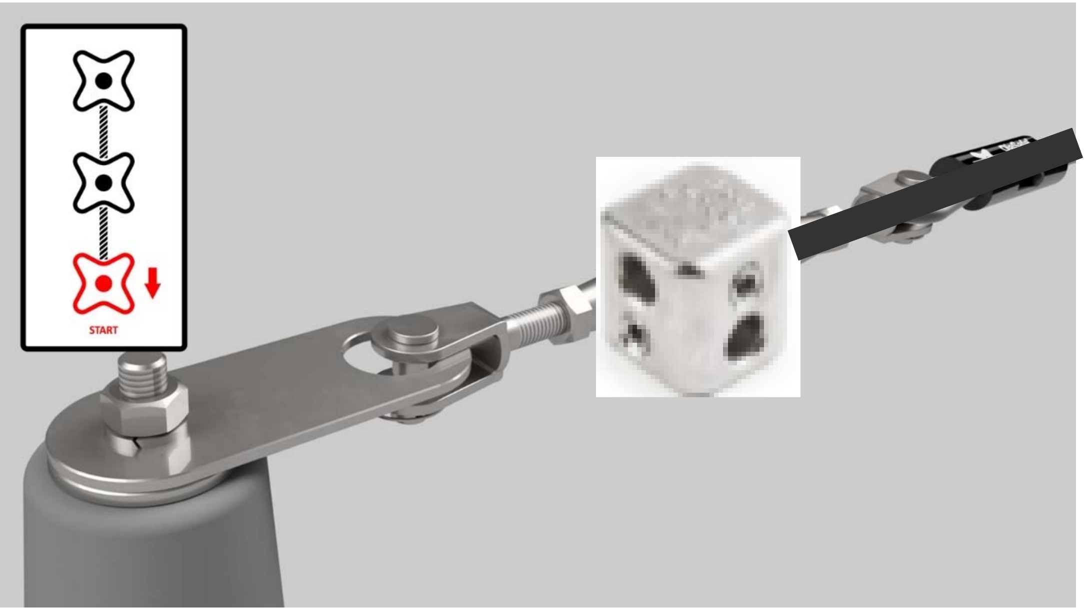


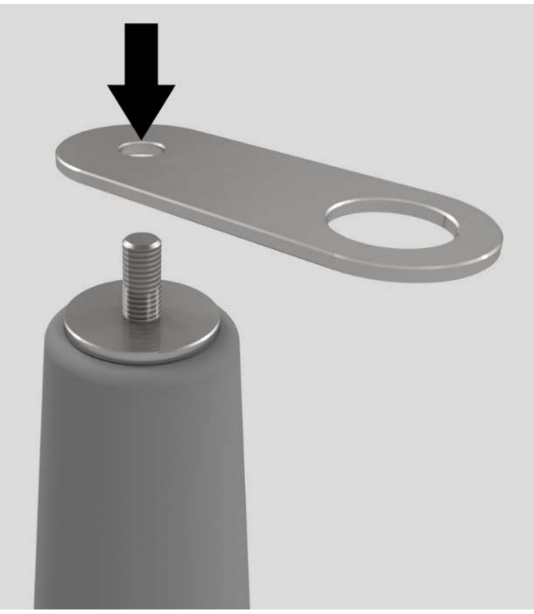

min. 12 cm

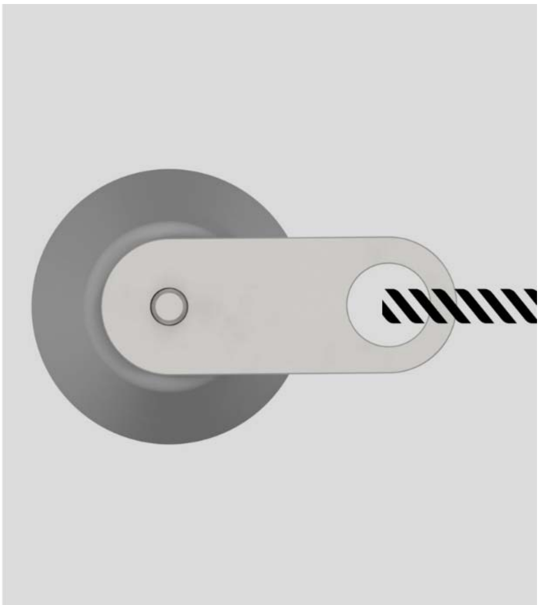

15
Nm
5
min. 12 cm
ElaSafe® Glide

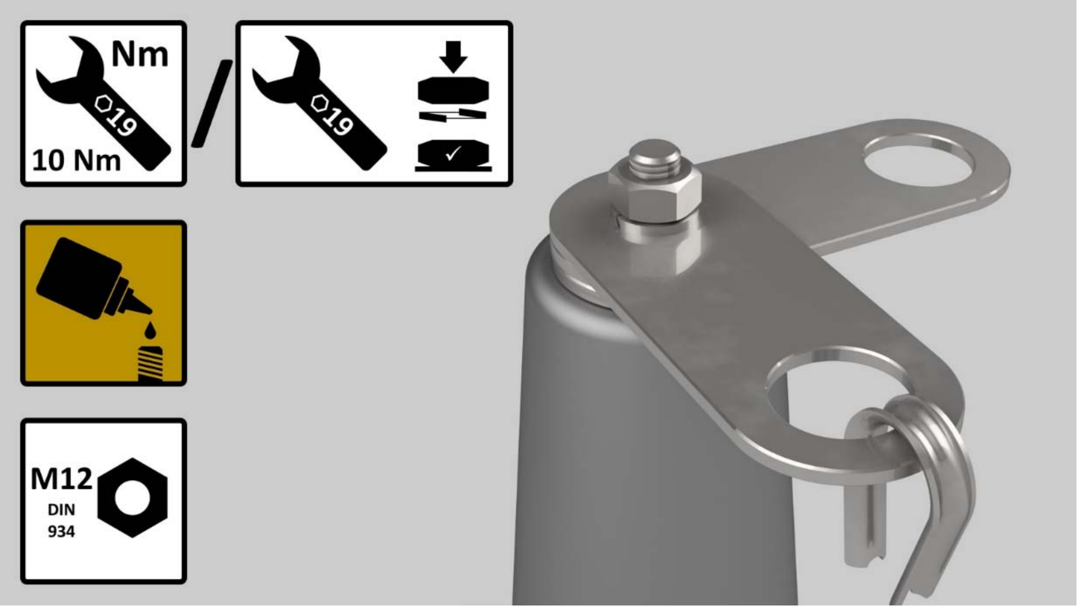

ElaSafe® Glide

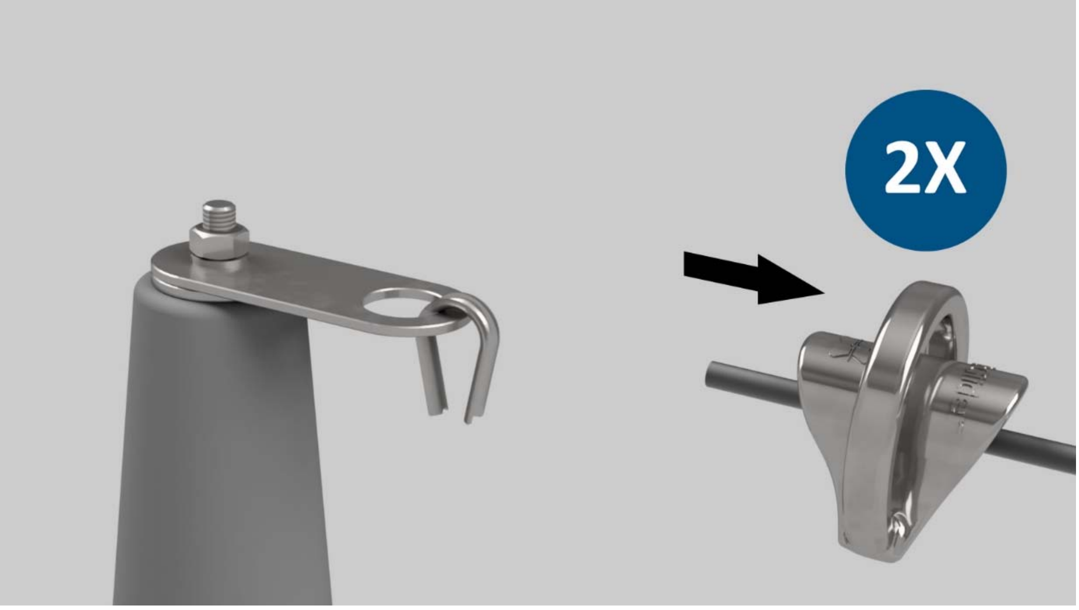

ClaSafe® Glide

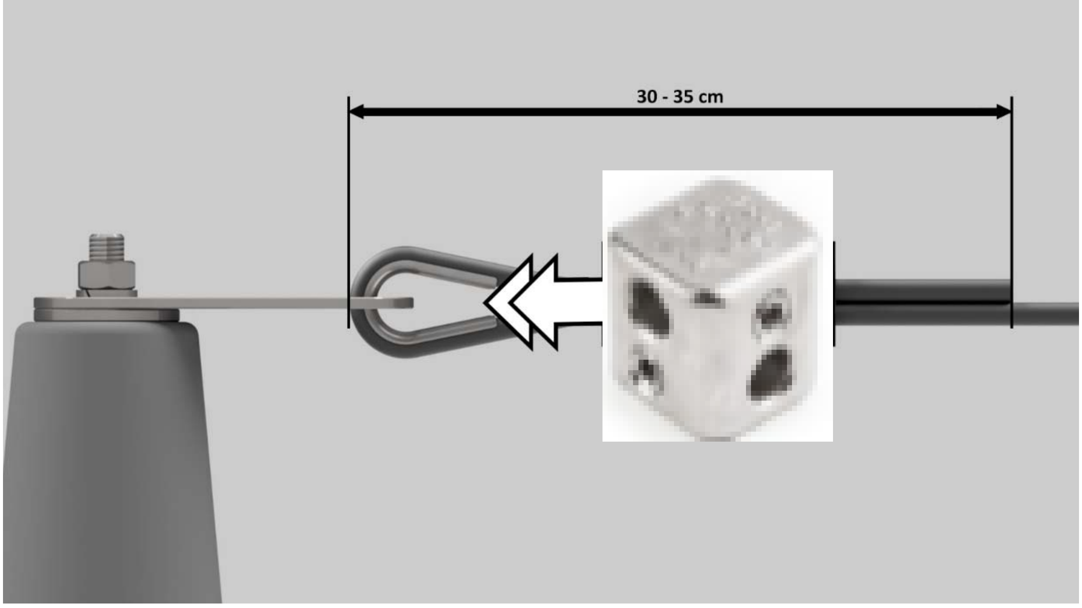

START





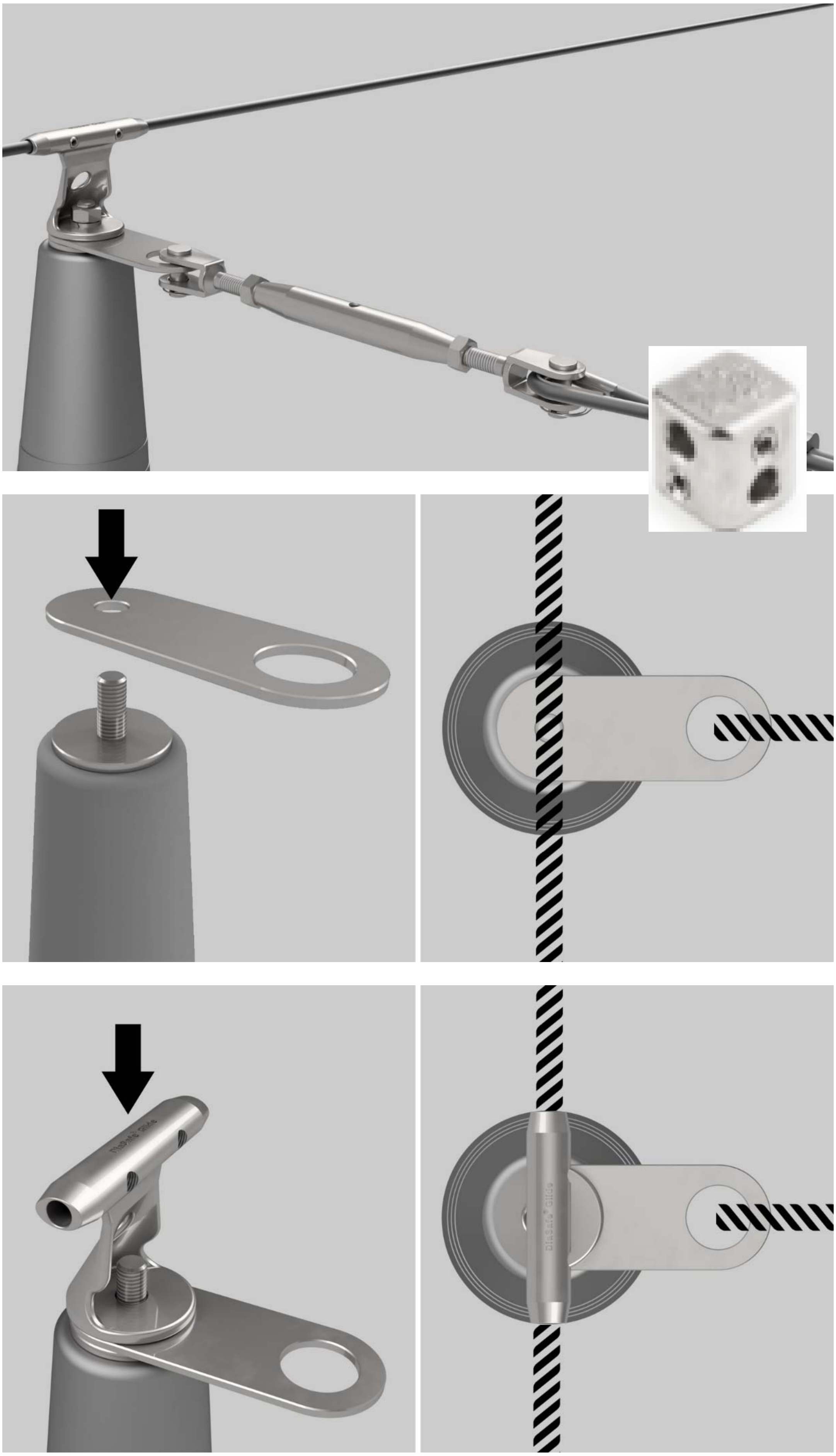

Nm
10 Nm
M12
DIN
934


2X

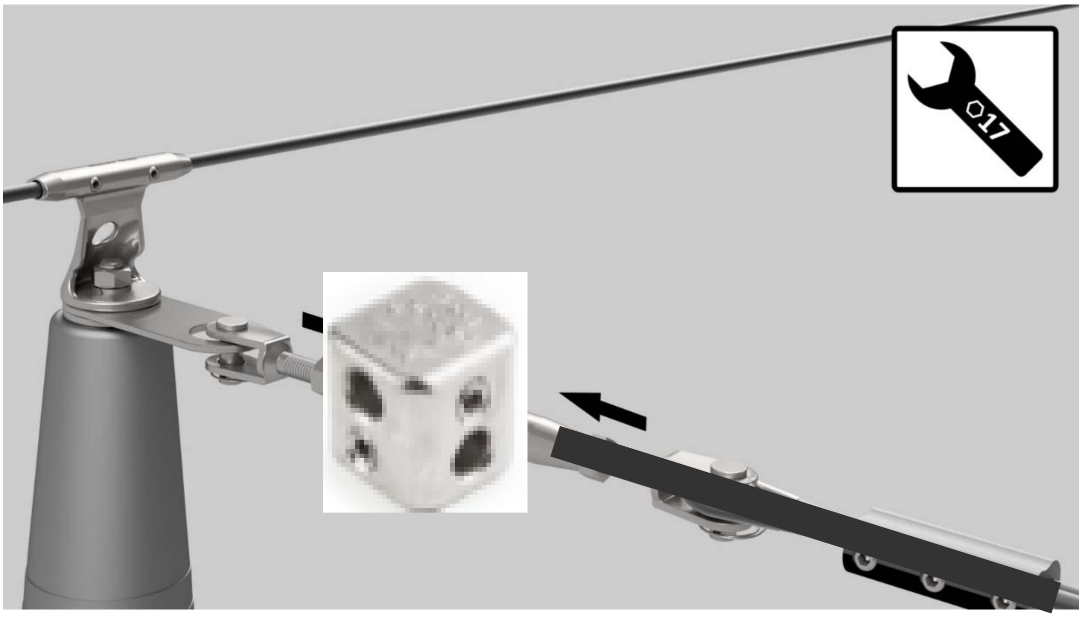

30 - 35 cm

OPTIONAL

T-connection / T-Verbindung / T-connection / incrocio a T / „T” elágazás

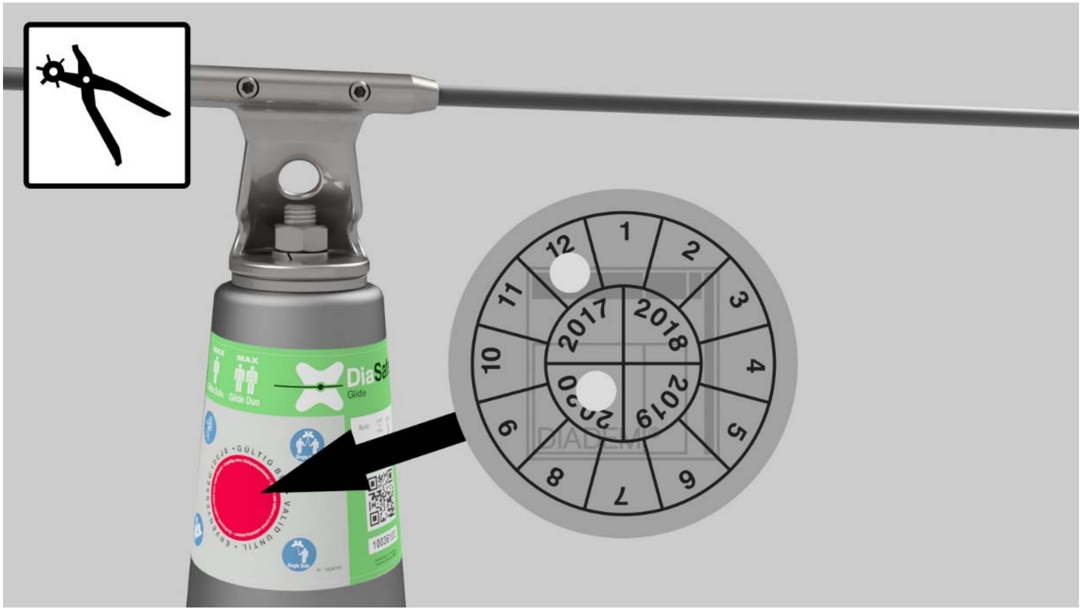


DiaSafe Glide
M12
DIN
127
19
19
Nm
10 Nm
M12
DIN
934



17

Filling the control label

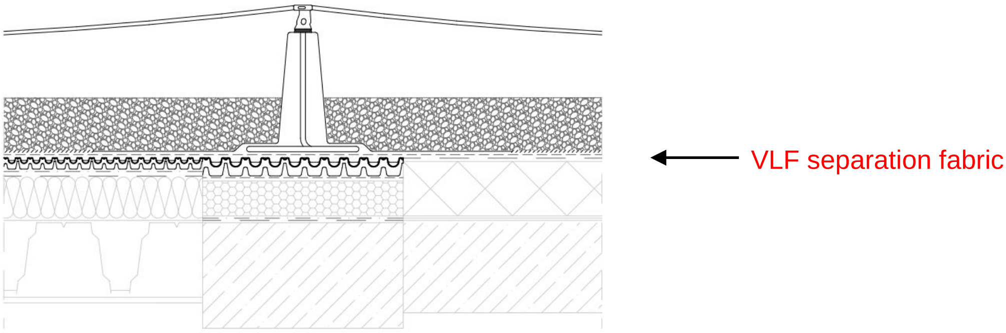


Use according to Technical Manual

6 Load bearing structure and ballast layer

6.1 Load bearing structure

DiaSafe® systems were tested and examined on a number of surface types and structures (roofs with steel, reinforced concrete and wooden structure). Their performance is controlled on the most prevailing waterproofing materials (bituminous, PVC, TPO or EPDM) and roof panes. The system is applicable on any roof system which is able to bear the extra load originating from the installation and use of the system.



Warning!

The system cannot be installed on substructure with uncompressed grained or rolling structure (e.g.: gravel or planting medium).

6.2 Ballast material

The stability of the system is ensured by the ballast layer which can be a planting medium for a green roof, otherwise gravel surfacing or other bulk material. DiaSafe® anchoring points can be used on roofs with a maximum 5° slope angle.

It shall be ensured that the surface weight of the ballast layer in a dry state on the entire surface of the ballasting mat:

by 1+1 persons:

- **should be 80 kg/m² at least when applying DS amoeba-shaped damping plate (3x3 m)**
- **or minimum 720 kg per post**
- **thickness of the ballast layer is minimum 3 cm in any case**

NUMBER OF USERS	**1+1**
standard mat size	**3 x 3 m**
surface weight	**80 kg/m²**
total weight per post	**720 kg**

Standard mat sizes belonging to the anchor points are 9,0 m^2 (3×3 m). The specified minimum ballast layer thickness of a minimum 3 cm shall always be ensured. When applying custom mat sizes, the minimum ballast surface weight shall be determined according to the Technical Manual to which assistance is provided by the table series below. The given mass is to be understood in dry condition!

1+1 users

Mat size	Total weight	Surface weight	Layer thickness: gravel, sand γ = 1600 kg / m³	Layer thickness: planting medium γ = 1000 kg / m³	Layer thickness: planting medium γ = 800 kg / m³
m^2 (m × m)	kg	kg / m^2	cm	cm	cm
4.0 (2 × 2)	720	180	10.5	18.0	22.5
6.0 (3 × 2)	720	120	7.0	12.0	15.0
9.0 (3 × 3)	720	80	5.0	8.0	10.0
12.0 (3 × 4)	720	60	min. 3.5	6.0	7.5
16.0 (4 × 4)	720	45	min. 3.0	4.5	6.0
20.0 (4 × 5)	800	40	min. 3.0	4.0	5.0
25.0 (5 × 5)	875	35	min. 3.0	3.5	4.0
30.0 (5 × 6)	900	30	min. 3.0	3.0	3.5
35.0 (5 × 7)	1050	30	min. 3.0	3.0	3.5
40.0 (5 × 8)	1200	30	min. 3.0	3.0	3.5

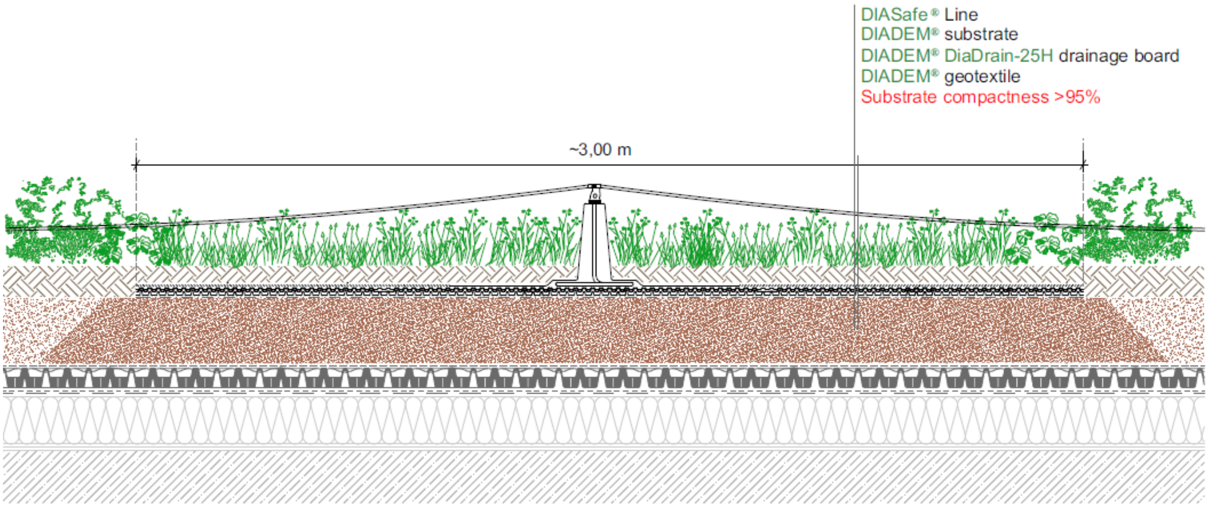
The given γ gross densities (specific weights) must be checked on site!

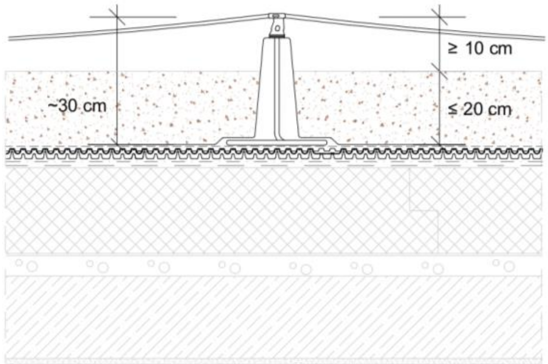
If the surface of the ballasting mat shall be increased, application of an auxiliary ballasting mat is required. If the surface of the ballasting mat shall be reduced for geometric reasons, the standard ballasting mat must be folded back to the appropriate size or cut, but in this case, the minimum distance of the mat measured from the centre point of the damping plate cannot be smaller than 50 cm at any point, and special attention shall be paid to the existence of the appropriate ballast weight. It is prohibited to cut or damage the glass-fibre reinforced plastic in any way.

Warning!

Because the thickness of the ballast layer may change over time (people walk on it, it is eroded by wind or rain etc.), the actual layer thickness must be controlled before each use by visual inspection at least. The ballast layer must always cover the ballasting mat on the entire surface. In case of inadequate layer thickness, additional ballast material is required.
The various types of ballast materials (substrate, gravel etc.) can be combined within one system; in such a case uniform weight distribution of the mixed ballast materials shall be ensured. This help is provided by the marking placed on the anchor post of the damping plate, which is described in detail in the Installation Guide.

6.2.1 Installation of the system in the case of ballast materials with various layer thicknesses





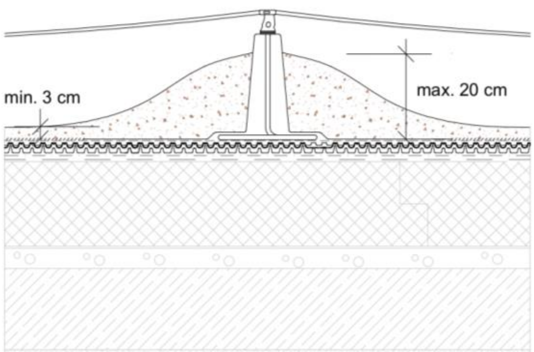
Layer thickness of the ballast materials on green roofs can be different. If thickness of the ballast layer is maximum 20 cm or less, the system can be installed in the regular way. For larger layer thickness, the system can be installed as shown in the picture above. This applies both for the wire-rope and single systems.

6.2.2 Uniform layer thickness

Basically, the uniform weight distribution of the ballast materials shall be ensured. Only the minimum layer thickness shall be ensured. The top of the post of the DiaSafe® systems shall be at least 10 cm above the surface of the ballast layer.

6.2.3 Varying layer thickness

Uneven ballast layer is permitted under observance of the load parameters.



7 Information on system installation and usage

- The system guarantees full-scale safety for the user, regardless of the sag of the wire.
- The sag of the wire may change during the lifetime of the system, e.g. due to the mounting actions, thermal expansion or other dynamic effects. It is important that the DiaSafe® systems are not preloaded systems, the wire does not need to be entirely tight, on the contrary, overtight wires originating from the incorrectly adjusted sag of the wire disadvantageously affect the efficiency and durability of the anchoring system.
- If posts of the system are distorted due to the incorrect wire tension during or after the installation, it means that the system was overstretched.
- Such bending of a post due to mounting, thermal expansion or other dynamic effects means solely an aesthetic change to the system, it cannot be subjected to manufacturer’s warranty procedure.
- The system is capable to accomplish its function even in the above cases.
- When using according to the respective intended use, the fixing element of post head can be used - released and then fixed again - safely so many times on the occasion of the mandatory check before maintenance, inspection and use, until no tear of a filament on the wire can be experienced and the clamping bolt can be operated as intended.

8 System installation and annual inspection information

8.1 System installation and annual inspection

- For the commissioning of the system, the Service book and the handover protocol shall be completed in compliance with the test criteria. The validating sticker shall be placed on the control label.
- The annual inspection shall be documented in writing. The test criteria and detailed information are included in the Service book. Based on the international guidelines and the manufacturer’s instructions, the inspection shall be performed without a test load.

8.2 Information regarding required free fall height

To appropriately fulfil the fall arresting function of the system it is required to consider the correct free fall height both for planning and before being put into service. To consider this, assistance is provided by the respective existing provisions.

Warning!

The system shall not provide a fall arrest function if the free fall height is below the min. 6,25 m. Displacement of the anchorage point and elongation of the safety rope must be taken into account in all cases.

9 Documentation

The manufacturer provides documentation for each **DiaSafe®** system attached and in digital, downloadable form. The installed falling arrest system can be registered on the **DIADEM® Online** (reg.diadem.com) registration interface. The Installation protocol is prepared during registration.

Parts of the documentation:

- Technical Manual (printed or downloadable)
- Installation Guide (printed or downloadable)
- Service Manual (furnished with individual serial number): (printed)
    - Handover protocol
    - Checking protocol
    - Validating decal
- Control label (printed)

At the annual inspection, the expert performing the inspection is obliged to place the control label validating the appropriate state of the installed fall arresting system on the control label of the system.

Warning!

In lack of a validly filled and logged Service Manual and/or Online System Registration the state of the system becomes uncontrolled and its functionality becomes uncontrollable. This completely excludes the Manufacturer's responsibility for eventual damages, faults or injuries.

10 Technical data

Maximum forces and displacements (Temperature: 20 °C):

System	Type	Test	Deflection [mm]	Max. Force Post [kN]	System build-up
DiaSafe®	Line	Dynamic	2105	5,096 / 6,405	Line (8m LINE)
DiaSafe®	Line-Glide	Dynamic	2497	5,60	combined with **RoofX®-C Glide** (8m LINE)
DiaSafe®	Line	Static	-	13,76	Line

Sufficient clearance under the usage area shall be ensured in any case! Depending on the length of wire the displacement may highly deviate from the data specified by the manufacturer.

11 Disposal

Do not dispose of the fall protection system in the house waste. Following national requirements, gather the used parts and dispose of them in an environmentally responsible fashion.